

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	DermaXplain - XAI for Reliable Healthcare Image Detection		
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Date of Submission:	09/11/2025		

Section 2: About your Practicum

What is the topic of your proposed practicum?

Our practicum project, *DermaXplain: Explainable AI (XAI) for Reliable Healthcare Image Detection*, applies XAI to skin cancer and lesion analysis. Traditional AI provides accurate but opaque diagnoses. Recent work tries to close the gap between accuracy and interpretability. Our goal is to evaluate XAI methods (Grad-CAM, SHAP, LIME) and define measurable explainability by comparing their saliency maps. Expert lesion masks serve as references for metrics assessing how well saliency aligns with true lesion areas. For images without expert masks, pseudo-masks will be generated automatically using segmentation models such as the U-Net.

Please provide details of the papers you have read on this topic:

1. **Codella, N. C. F., et al. (2019).** *Skin Lesion Analysis Toward Melanoma Detection: A Challenge Hosted by the International Skin Imaging Collaboration (ISIC 2018)*. Data in Brief, Volume 29, 105374.

This paper describes the ISIC 2018 challenge, including three tasks: lesion segmentation, attribute detection, and disease classification. Task 1 provides lesion masks computed or drawn, but all expert validated, which are the ground truth for segmentation and explainability evaluation.

2. **Tschandl, P., Rosendahl, C., & Kittler, H. (2018).** *The HAM10000 Dataset, a Large Collection of Multi-Source Dermatoscopic Images of Common Pigmented Skin Lesions*. Scientific Data, 5, 180161.

The HAM10000 dataset is widely used for skin lesion classification. It provides over 10,000 dermatoscopic images across seven diagnostic categories. We aim to pair it with ISIC 2018 for explainability and segmentation experiments.

3. **Yu, R., Zhang, L., Liu, Q., & Wang, H. (2024).** *Skin lesion image segmentation based on an improved U-Net*. *Applied Intelligence*. Vol. 8 - issue 3, 609-618.

The paper presents an improved U-Net architecture, a convolutional neural network designed for biomedical image segmentation. U-Net (Ronneberger et al., 2015) has become a standard for lesion boundary detection. It can be used to automatically generate pseudo-masks for datasets such as HAM10000.

4. **Saporta, A., et al. (2022).** *Benchmarking Saliency Methods for Chest X-ray Interpretation*. *Nature Machine Intelligence*, 4(9), 767–778.

Saporta et al. describe the usage of the Intersection over Union (IoU), Dice coefficient, and Saliency-Inside Ratio (SIR) measures for saliency in Chest X-ray interpretation. We will adapt this methodology to skin lesion interpretation.

5. **Srivastava, Kriti et al. (2024).** *Enhancing Interpretability, Reliability, and Trustworthiness: Applications of Explainable Artificial Intelligence in Medical Imaging, Financial Markets, and Sentiment Analysis*. *2024 16th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*

This paper reviews explainable AI methods like LIME, SHAP, and Grad-CAM, showing how they improve transparency, interpretability, and trust in AI applications across medical imaging, financial markets, and sentiment analysis. It finds that the choice of XAI technique depends on the data type and domain, and emphasises the need for ongoing research to tackle ethical, technical, and practical challenges.

How does your proposal relate to existing work on this topic described in these papers?

Recent work in medical imaging, with its high diagnostic accuracy, is facing demand for more transparency and interpretability from clinicians.

The **ISIC 2018** and **HAM10000** datasets (**Codella et al., 2019; Tschandl et al., 2018**) form the standard foundation for skin lesion research. It provides expert annotations and high-quality dermoscopic images.

Recent segmentation studies, such as **Yu et al. (2024)**, demonstrate that **improved U-Net** architectures can generate high-accuracy lesion masks, measured with IoU and Dice scores. This supports our use of U-Net to create **pseudo-masks** for evaluating explainability methods in dermatology.

Recently, **Srivastava, Kriti et al. (2024)** have applied **Grad-CAM, LIME**, and **SHAP** to visualise how models focus on diagnostic features. Their comparison, mostly qualitative, remains an important step in XAI.

We aim in our practicum to extend these studies by applying a quantitative evaluation of explainability methods in dermatology. Building on prior XAI benchmarks (**Saporta et al., 2022; Wang et al., 2023**), we adopt overlap-based metrics such as Intersection over Union (IoU), Dice coefficient, and Saliency-Inside Ratio (SIR) to measure alignment between saliency maps and lesion masks. By using expert masks from ISIC 2018 compared with pseudo-masks generated by the U-Net for HAM10000, our project connects traditional segmentation to explainability research. We aim to create a reproducible, quantitative benchmark for evaluating XAI methods in skin lesion classification.

What are the research questions that you will attempt to answer?

1- How well do XAI methods align with the clinically relevant regions?

We will quantify this alignment using expert masks from ISIC 2018 and pseudo-masks generated by U-Net for HAM10000, applying metrics such as Intersection over Union (IoU), Dice coefficient, and Saliency-Inside Ratio (SIR).

2- How stable are these XAI methods when faced with small changes in the input?

We will test how deletion or progression insertion of important areas to image features affects the corresponding saliency maps. A stability analysis under small perturbations (rotation, noise, brightness) will also be carried out.

3- Is it possible to generate pseudo-masks of similar quality to the expert annotations for explainability evaluations?

We will compare results obtained using expert and U-Net-generated masks to assess whether pseudo-masks can provide reliable ground truth in the absence of clinical annotations.

How will you explore these questions?

- What software and programming environment will you use?

We will use Python in Jupyter Notebook or Google Colab, with deep learning frameworks like TensorFlow or PyTorch for model implementation and libraries such as OpenCV, NumPy, and Matplotlib for image analysis and visualization.

- What coding/development will you do?

We plan to code the segmentation bit using the U-Net architecture, apply the XAI methods (Grad-CAM, LIME, SHAP) to skin lesion images, and implement scripts to quantitatively evaluate explainability results against expert and pseudo-masks.

- What data will be used for your investigations?

The ISIC 2018 dataset (with expert-annotated masks) and HAM10000 dataset (with U-Net-generated pseudo-masks) will be used for all segmentation, classification, and explanation experiments.

- Is this data currently available? If not, where will it come from?

Both datasets are publicly available from their respective sources online. ISIC 2018 and HAM10000 are hosted by the International Skin Imaging Collaboration (ISIC) Archive (<https://www.isic-archive.com/>).

- What experiments do you expect to run?

We expect to run segmentation with U-Net, generate and compare saliency maps (using XAI methods), and quantitatively measure how well these explanation maps align with lesion masks.

- What output do you expect to gather?

We will expect to gather quantitative alignment scores for each XAI method, visual comparisons of saliency maps versus ground-truth masks, and reproducible benchmark scripts for evaluation.

- How will the results be evaluated?

We use IoU values between the three XAI models to compare the overlap between the masks and the saliency. Other accuracy scores, such as Dice coefficient and Saliency-Inside Ratio (SIR), will also be generated for evaluation.